

INTERNATIONAL STANDARD



Digital video interface – Gigabit video interface for multimedia systems



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Digital video interface – Gigabit video interface for multimedia systems

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**DIGITAL VIDEO INTERFACE –
GIGABIT VIDEO INTERFACE FOR MULTIMEDIA SYSTEMS**

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The text of this standard is based on the following documents:

CDV	Report on voting
100/2193/CDV	100/2298/RVC

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INTRODUCTION

This International Standard is based on a standard JEITA CP-6101: Digital monitor interface GVIF that was originally specified by the Japan Electronics and Information Technology Industries Association (JEITA).

The gigabit video interface (GVIF) is a serial point to point interface supporting uncompressed digital video links that was designed to address the needs of automotive navigation and entertainment systems, etc., to transport base band digital video information. The GVIF applies low voltage differential signaling (LVDS) technology and makes use of a thin cable consisting of a single shielded twisted pair of conductors that exhibits high noise immunity and low EMI, and is optimized for small size and low weight. The GVIF supports display resolutions ranging from WQVGA through WUXGA with maximum 24 bit per pixel colour video data, and can transmit base band video signal over cable lengths over 10 m. When paired with high bandwidth data content protection (HDCP), the GVIF's standard functions and features address all of the requirements for delivering content protected video from a source to a video display monitor. Optionally, the GVIF supports audio data transmission and user data transmission.

The Association of Radio Industry Business (ARIB) refers the GVIF in its standard ARIB STD-B21 as one of authorized digital video output interfaces.

DIGITAL VIDEO INTERFACE – GIGABIT VIDEO INTERFACE FOR MULTIMEDIA SYSTEMS

1 Scope

This International Standard describes a serial digital interface, gigabit video interface (GVIF) for the interconnection of digital video equipment. The GVIF is primarily intended to carry high-speed digital video data for general usage and is well suited for multimedia entertainment systems in a vehicle.

This International Standard specifies the physical layer of the interface including transmission line characteristics and electrical characteristics of transmitter and receiver. Mechanical and physical specifications of connectors are not included.

2 Normative references

The following documents, in whole or in part, are normatively referenced in this document and are indispensable for its application. For dated references, only the edition cited applies. For undated references, the latest edition of the referenced document (including any amendments) applies.

IEC 62315-1:2003, *DTV profiles for uncompressed digital video interfaces – Part 1: General*

ITU-R BT.601-5, *Studio encoding parameters of digital television for standard 4:3 and wide-screen 16:9 aspect ratios*

ITU-R BT.656-5, *Interface for digital component video signals in 525-line and 625-line television systems operating at the 4:2:2 level of Recommendation ITU-R BT.601*

3 Terms, definitions and abbreviations

3.1 Terms and definitions

For the purposes of this document, the following terms and definitions apply.

3.1.1

DE

display enable signal given in IEC 62315-1

3.1.2

HSYNC

display horizontal synchronous signal given in IEC 62315-1

3.1.3

VSYNC

display vertical synchronous signal given in IEC 62315-1

3.1.4

RGB

display red, green, blue colour data input (TX) or output (RX) given in ITU-R BT.601-5 and ITU-R BT.656-5

3.1.5

YU(Cb)V(Cr)

display Y, U (Cb), V (Cr) pixel data input (TX) or output (RX) given in ITU-R BT.601-5 and ITU-R BT.656-5

3.1.6

CNTL/AUX

down-stream user defined signal or audio enable signal

3.1.7

P[23:0]

digital signal data like a 24 bit colour video data such as RGB or YU (Cb) V (Cr) data input (TX) or output (RX)

3.1.8

GVIF RX

circuit that receives the serial signal from a shielded-pair transmission line, decodes them and outputs to convert into the parallel video signal

3.1.9

GVIF TX

circuit that receives the parallel video signal, the control signals, and encodes them into serial data to send a signal by driving a shielded-pair transmission line

3.1.10

LOS

loss of signal

detection signal, asserted when the differential input signal at the receiver cannot receive

3.1.11

RX front-end

front-end block of receiver side

3.1.12

SDA

serial data

down-stream signal

3.1.13

SDATAP

down-stream positive-phase side signal of the differential serial data

3.1.14

SDATAN

down-stream negative-phase side signal of the differential serial data

3.1.15

REFRQP

current source signal for reference clock request from Rx side

3.1.16

REFRQN

current source signal for reference clock request from Rx side as well as REFRQP

3.1.17

SFTCLK

pixel clock

clock for capture of the parallel video data per pixel

3.1.18**TDA**

transmit data

down-stream user defined signal

3.1.19**TX front-end**

front-end block of transmitter side

3.1.20**UDA**

user data

up-stream user defined signal

3.1.21**IRQ**

up-stream common-mode reference request current for REFRQP/N

3.1.22**VOS**

common-mode voltage amplitude of reference request

3.1.23**VOD**

differential voltage amplitude for SDATAP/N

3.1.24**VDD**

power supply on the transmitter side

3.1.25**V_SDATAP**

single-ended voltage of SDATAP

3.1.26**V_SDATAN**

single-ended voltage of SDATAN

3.1.27**TP1**

transmitter end point for eye mask specification

3.1.28**normalized differential voltage**

voltage of transmitter output point

3.1.29**UI**

normalized time unit interval of transmitter output point

3.2 Abbreviations

AC	Alternating Current
DC	Direct Current
EMI	Electro-Magnetic Interference
GVIF	Gigabit Video InterFace
LSB	Least Significant Bit

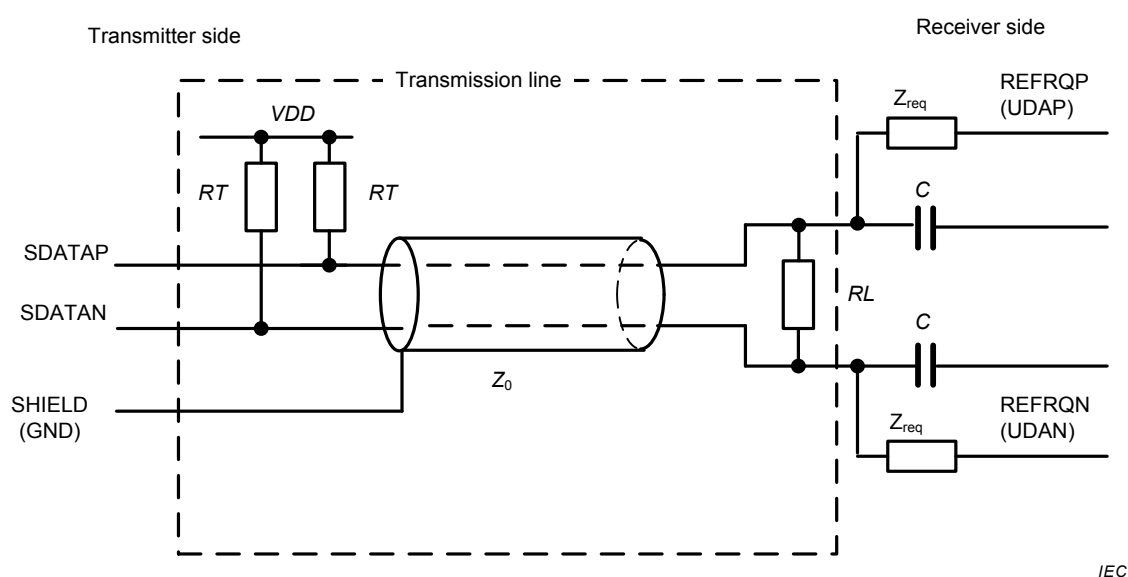
LVDS Low Voltage Differential Signaling

MSB Most Significant Bit

4 Architecture

Figure 1 illustrates the architecture of the GVIF. The fundamental operation of the GVIF is a simultaneous bi-directional data transmission technology, in which the low voltage differential signal is transmitted down from the transmitter side to the receiver side, and the common-mode voltage signal is transmitted up from the receiver side to the transmitter side through a shielded twisted differential pair cable.

The shielded twisted pair transmission line has the characteristic impedance Z_0 (see Figure 10), the line is terminated to VDD by RT of $(50 \pm 15) \Omega$ on the transmitter side, and is terminated carrying differential data in RL of $(100 \pm 5) \Omega$ on the receiver side.



where

RT are the pull-up terminated load resistors on the transmitter side $(50 \pm 15) \Omega$;

Z_0 is the characteristic impedance of the shielded twisted pair transmission line;

RL is the terminated resistor between differential data lines on the receiver side $(100 \pm 5) \Omega$;

C are AC coupling capacitors.

SDATAP/SDATAN are the down-stream positive and negative phases side signals carrying differential serial data.

REFRQP (UDAP)/REFRQN (UDAN) is the up-stream REFREQ common-mode current signal or UDA common-mode current user defined data signal. UDAP/UDAN are optional.

SHIELD (GND) is the GND and shielded ground for cable.

Z_{req} is a blocking filter for the up-stream signal. It can use resistors or inductors depending on the system implementation.

Figure 1 – Architecture of the GVIF

5 Electrical characteristics

5.1 DC electrical specifications

The DC electrical specifications of the transmitter side are shown in Table 1, and the DC electrical specifications of the receiver side are shown in Table 2.

Table 1 – DC electrical specifications of the transmitter

	Differential output peak to peak voltage (SDATAP/N)	Common mode voltage (SDATAP/N)		Input REFRQ assert current (SDATAP/N)	Input REFRQ de-assert current (SDATAP/N)
	mV	V		mA	mA
	Condition: $R_T = 50 \, \Omega$ $R_L = 100 \, \Omega$	Condition: $R_T = 50 \, \Omega$ $R_L = 100 \, \Omega$ $I_{RQ} = 0 \, \text{mA}$	Condition: $R_T = 50 \, \Omega$ $R_L = 100 \, \Omega$ $I_{RQ} = 11 \, \text{mA}$		
Minimum	690	$V_{DD} - 0,55$	$V_{DD} - 1,2$		-2,0
Typical	800				
Maximum	910	$V_{DD} - 0,35$	$V_{DD} - 0,8$	-7,3	

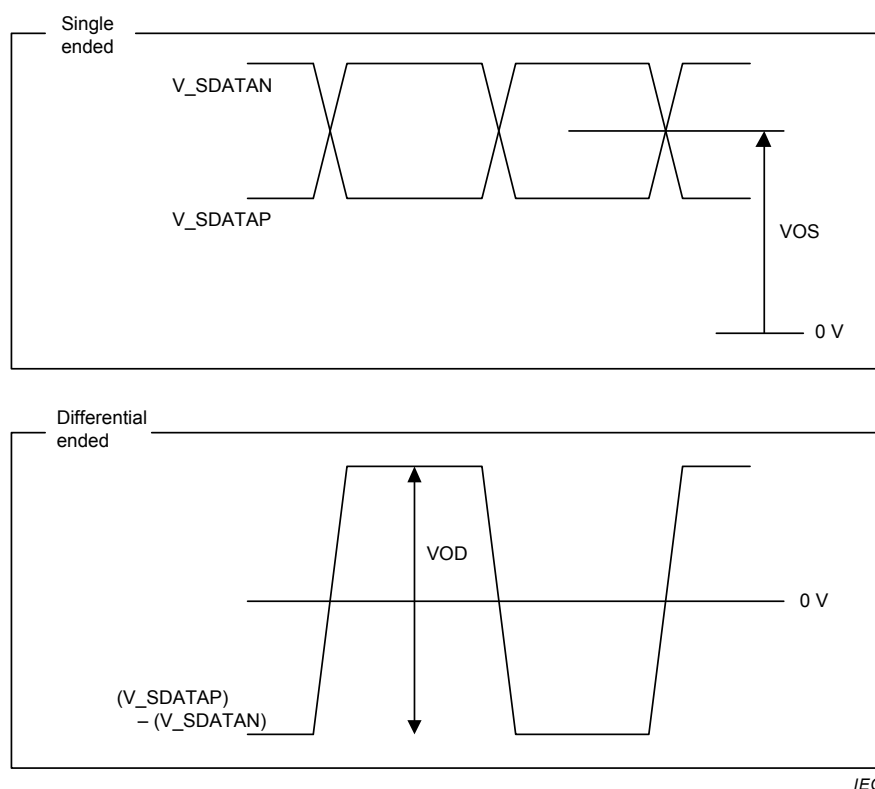


Figure 2 – VOD, VOS diagram

Table 2 – DC electrical specifications of the receiver

	Output HIGH current (REFRQP/N) mA	Output LOW current (REFRQP/N) mA
Minimum	–0,1	7,4
Maximum	0,1	11

5.2 AC electrical specifications

The AC electrical specifications of the transmitter side are shown in Table 3 and Figure 3 shows a transmitter end point eye specification (TP1). The AC electrical specifications of the receiver side are shown in Table 4.

Table 3 – AC electrical specifications of the transmitter

	SFTCLK frequency MHz	UDA data rate (up-stream) Mbit/s	SFTCLK duty factor %
Minimum	7,6	0,01	40
Maximum	160	2,41	60

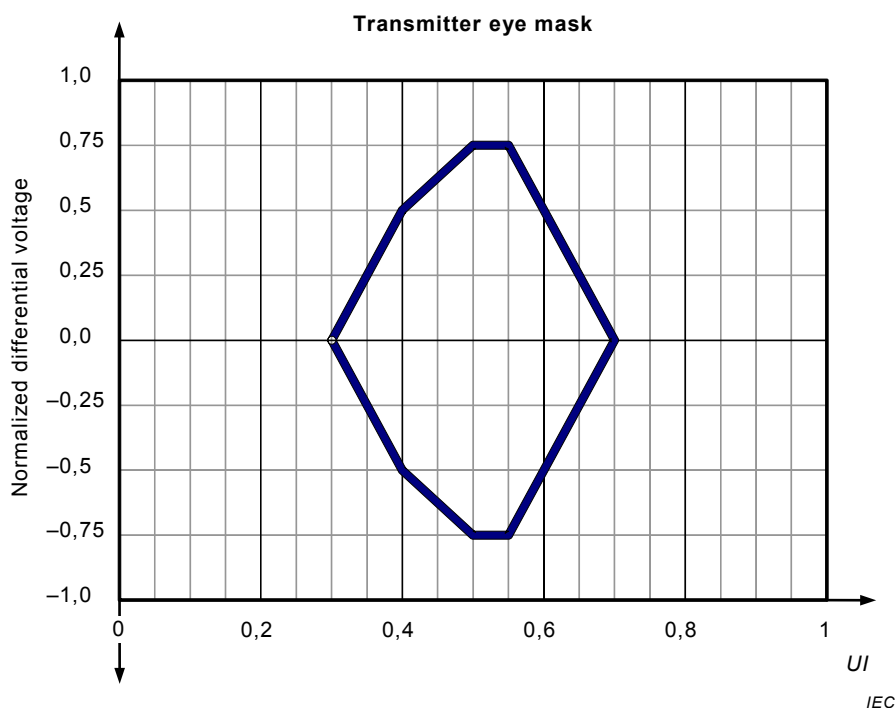


Figure 3 – Transmitter eye mask specifications (TP1)

Table 4 – AC electrical specifications of the receiver

	SFTCLK frequency MHz	UDA data rate (up-stream) Mbit/s
Minimum	7,6	0,01
Maximum	160	2,41

6 Front-end

6.1 General

The front-end block diagram of GVIF is shown in Figure 4.

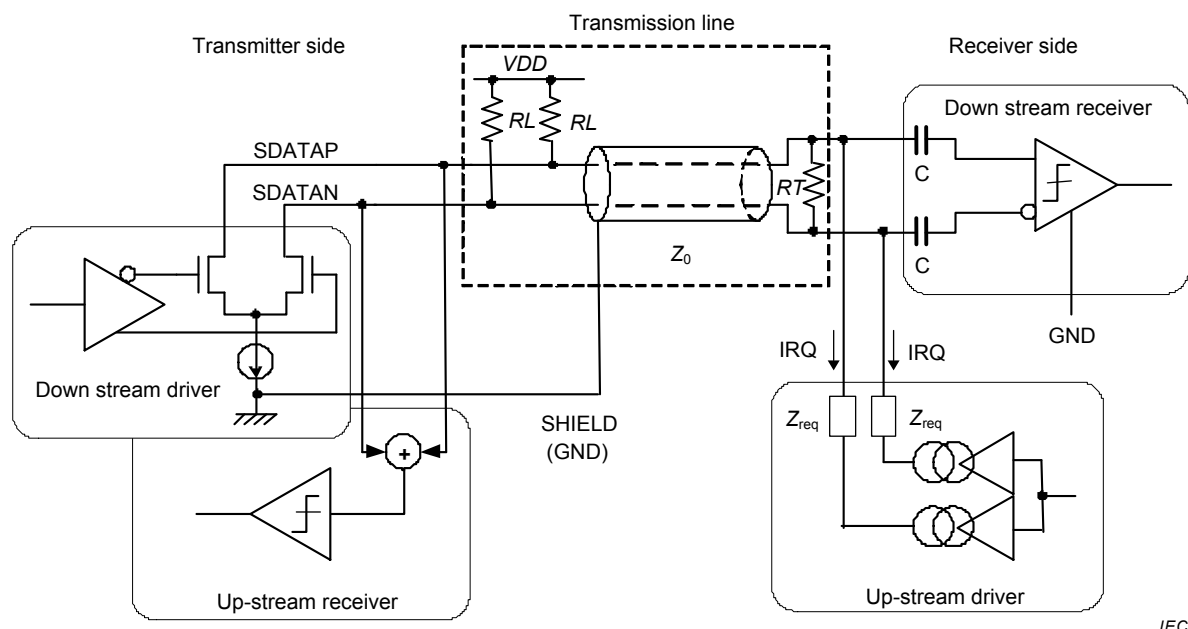


Figure 4 – Front-end block diagram

6.2 TX front-end

The TX front-end consists of a termination circuit, a down-stream driver and an up-stream receiver. The termination circuit consists of 2 resistors R_L , and the SDATAP/N differential signal is pulled up to voltage reference (VDD) with a $(50 \pm 15) \Omega$ resistor. The down-stream driver consists of a differential current output circuit that is driven by the serial signal from the encoder. The up-stream receiver detects the common-mode signal which RX sends through the shielded twisted pair line. The input to the down-stream driver has two modes. One is the serialized actual encoded video data input mode and the other is the reference clock signal for REFREQ hand-shake input mode. These two modes activate depending on the common-mode signal level. The common-mode signal level is normally high. When a long low level pulse is detected, the up-stream receiver activates the REFREQ signal, and changes a mode of the encoder into the reference clock mode. In case of the optional up-stream user data transmission, the up-stream receiver outputs the common-mode voltage as an UDA signal by using binary digital data sent to the encoder. In this case, the upper limit of the low pulse time is 100 μ s.

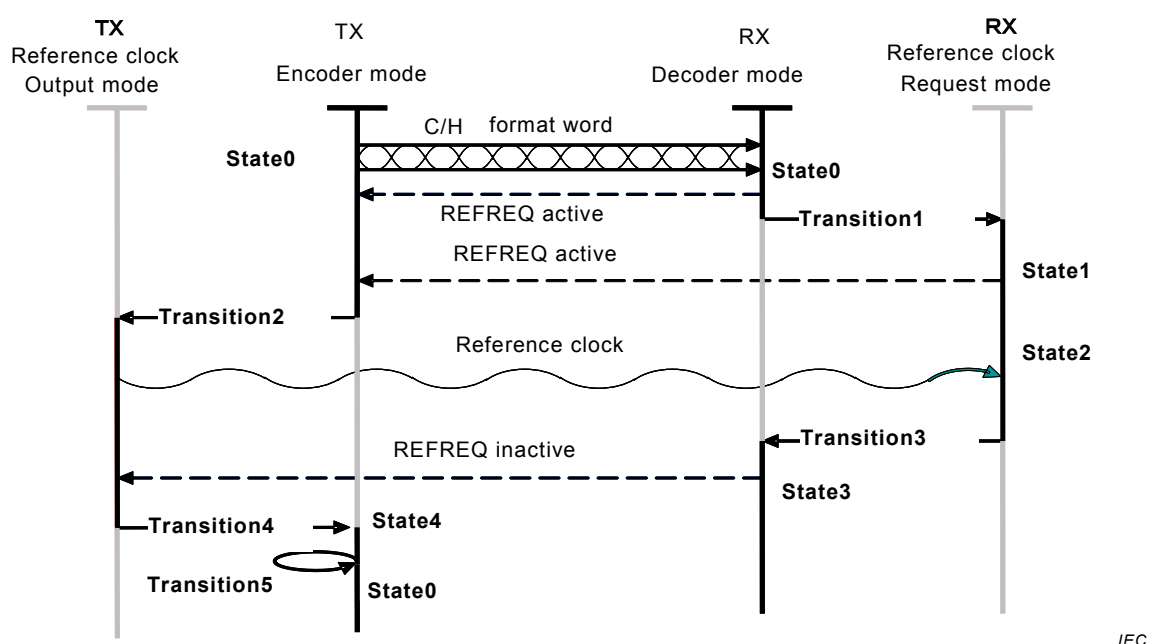
6.3 RX front-end

The Rx front-end consists of AC capacitors, a termination resistor R_T ($100 \pm 5 \Omega$), a down-stream receiver and an up-stream driver. The down-stream receiver consists of a differential input detection circuit which receives the transmission potential differential signal through the shielded twisted pair line. The up-stream driver drives the up-stream transmission signal applying a current through the termination resistor Rx through the shielded twisted pair transmission line. (A recommended transmission system and transmission line for electrical characteristics is specified in Clause 5.)

7 Transition state link

The transition state link of GVIF shall meet the procedure described below.

There are two states in the connection link between GVIF TX and GVIF RX. One is the state transmitting differential signal with a reference clock, the other is the state transmitting the H format word or the C format word. In the former state, the TX encoder is in the reference clock output mode and the RX decoder is in the reference clock request mode. In the later state, the TX encoder changes into the encoder mode and the RX decoder changes into the decoder mode. The state transition switching diagram of the encoder and the decoder is shown in the Figure 5.



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State0 (normal)	:	The C/H format word is transmitted down from the TX in the encoder mode to the RX, and the deactivation signal REFREQ is transmitted p from the RX in the decoder mode to the TX.
Transition1	:	Transition to the reference clock request mode after finding an irregular HSYNC when the RX decodes.
State1	:	The RX transmits up the activate signal REFREQ.
Transition2	:	The TX transits to the reference clock output mode when the activate signal REFREQ is detected.
State2	:	The TX transmits down the reference clock, and the RX adjusts the internal sampling clock.
Transition3	:	The RX transits to the decoder mode after the internal sampling clock adjustment.
State3	:	The RX transmits up the inactivate signal REFREQ.
Transition4	:	The TX transits to the encoder mode when the inactivate signal REFREQ is detected.
State4	:	P[23:0] transmits continuously the H/C format word equivalent all zero until the TX transmits (VSYNC, HSYNC) (1,1) → (1,0) 60 times.
Transition5	:	Return to normal when the signal has been transmitted 60 times.

Figure 5 – Transition state link

8 Protocol

8.1 General

The encoder encodes the 30 bit of data (P[23:0], HSYNC, VSYNC, DE, CNTL, SDA and TDA) in synchronization with the input of SFTCLK, and outputs 1 bit of the serial signal S to the TX front-end.

To ensure the DC balance data and a reasonable transition, it is required to generate a synchronization pattern for each word in synchronization with the falling edge of HSYNC at the receiver.

8.2 Encoder

The encoder encodes the full 30 bit of input data (P[23:0], HSYNC, VSYNC, DE, CNTL, SDA and TDA) synchronized with SFTCLK, and outputs a 1 bit serial signal S to the TX front-end. The signal is coded after dividing into the following data.

- Broadband data P[23:0] (24 bits), no transition data.
- Time mark data HSYNC, VSYNC, DE, CNTL, SDA and TDA (6 bit).

Transition frequency of the signal is limited by the logical specification coding.

The broadband data are normally converted to 1 bit data, but in case of the time mark data, the transition is converted to 1 bit data. When there is no transition in the time mark data the broadband data are converted to the C format with 20 % overhead. When there is a time mark transition, the broadband data are converted to the H format with 6 bit header and 24 bit broadband data.

The broadband data and the time mark data are output as a serial signal S led by the MSB after conversion into a 30 bit length C format word or H format word.

The C format word is used when there is no time mark data transition at the previous pixel clock cycle, and the H format word is used when there is/are one or more time mark data transition(s) at the previous pixel clock cycle. (See Figure 6).

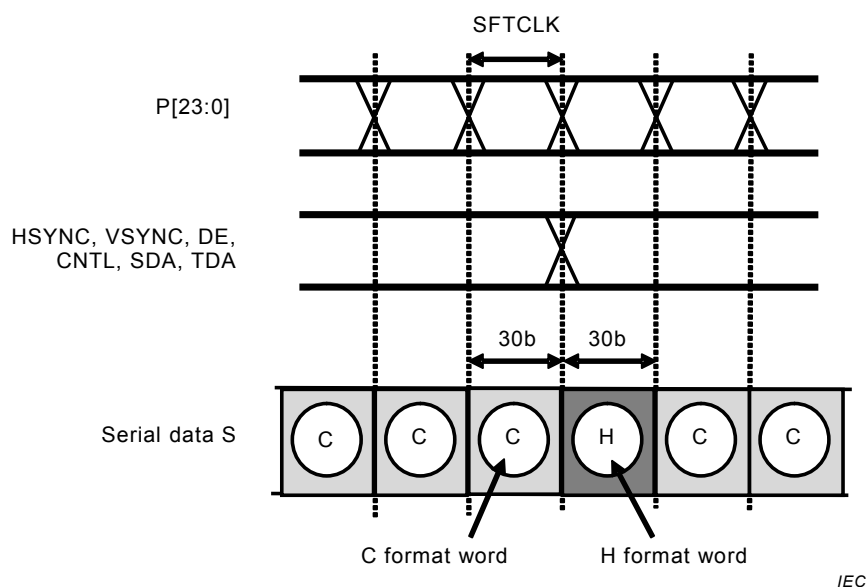


Figure 6 – Encoder output diagram

The C format word consists of the combined six codes of 5 bit which is generated by the 4B5B conversion breaking the broadband data P[23:0] by 4bit. (See Figure 7).

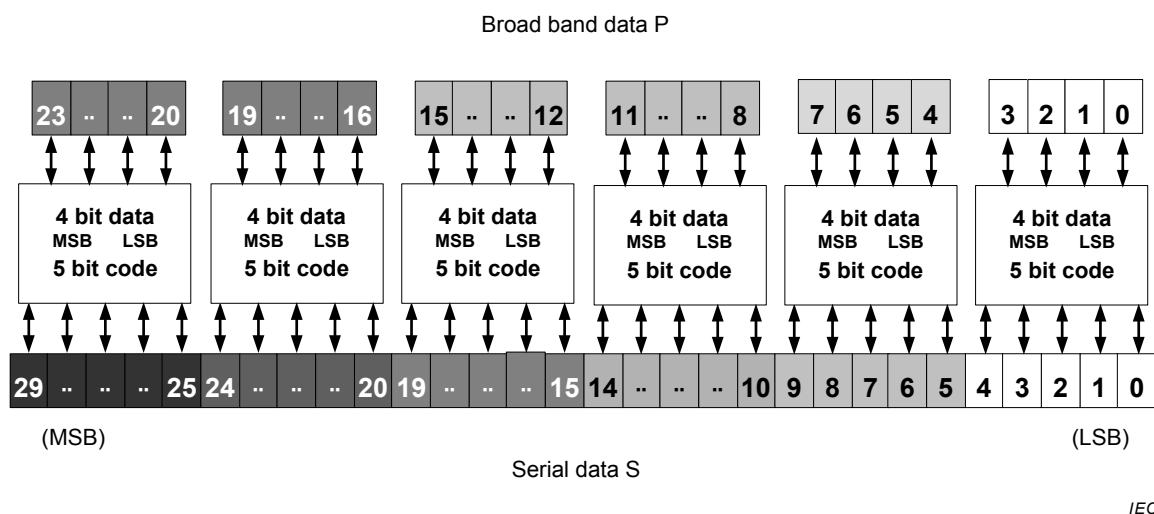


Figure 7 – C format word

Table 5 – 4B5B conversion

4 bit data MSB – LSB	5 bit code MSB – LSB	4 bit data MSB – LSB	5 bit code MSB – LSB
"0 0 0 0"	"0 0 1 0 1"	"1 0 0 0"	"1 0 0 1 0"
"0 0 0 1"	"0 0 1 1 0"	"1 0 0 1"	"1 0 0 1 1"
"0 0 1 0"	"0 0 1 1 1"	"1 0 1 0"	"1 0 1 0 0"
"0 0 1 1"	"0 1 0 0 1"	"1 0 1 1"	"1 0 1 0 1"
"0 1 0 0"	"0 1 0 1 0"	"1 1 0 0"	"1 0 1 1 0"
"0 1 0 1"	"0 1 0 1 1"	"1 1 0 1"	"1 1 0 0 1"
"0 1 1 0"	"0 1 1 0 0"	"1 1 1 0"	"1 1 0 1 0"
"0 1 1 1"	"0 1 1 0 1"	"1 1 1 1"	"1 1 1 0 0"

The H format is generated by a combination of the 24 bit broadband data P[23:0] with a 6 bit header that indicates the transition state of a time mark, see Figure 8. The positions of even numbers of the broadband data P are inverted in the serial data S. The structure of the header is shown in Table 6.

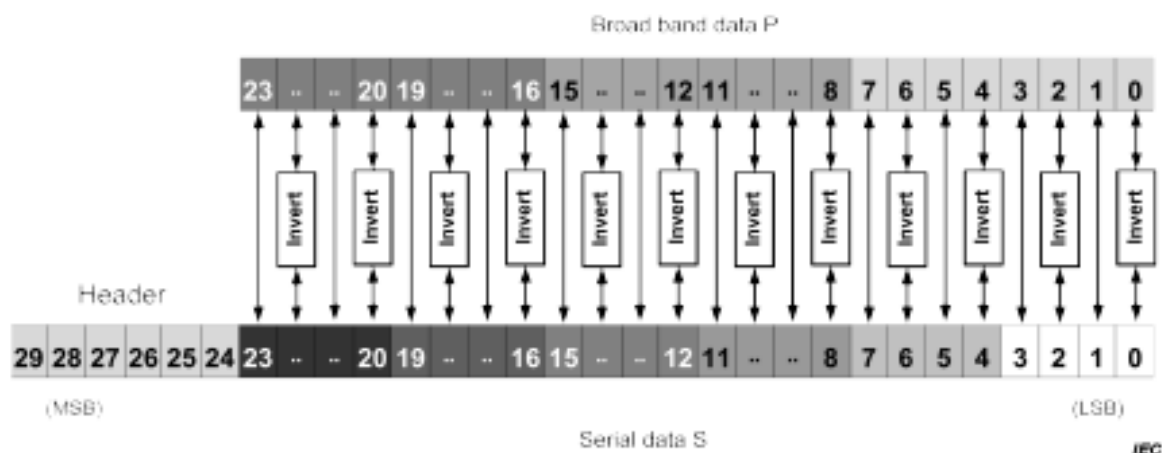


Figure 8 – H format word

Table 6 – VSYNC, HSYNC, DE, CNTL/AUX, SDA, TDA transition and the corresponding header

	Transition signal	Header bit array	Remark
a	VSYNC, HSYNC	"1 0 0 0 V H"	V and H are the VSYNC inversion value and the HSYNC value after transition.
b	DE, CNTL/AUX	"0 1 1 1 D C"	D and C are the DE and CNTL values after transition.
c	SDA, TDA	"1 1 1 1 S T"	S and T are the SDA and TDA values after transition.
Transition between the signals simultaneously among a, b and c shall not be permitted.			

8.3 Decoder

The serial data S that comes from the RX front-end is converted as shown in Figure 7, Figure 8, Table 5 and Table 6.

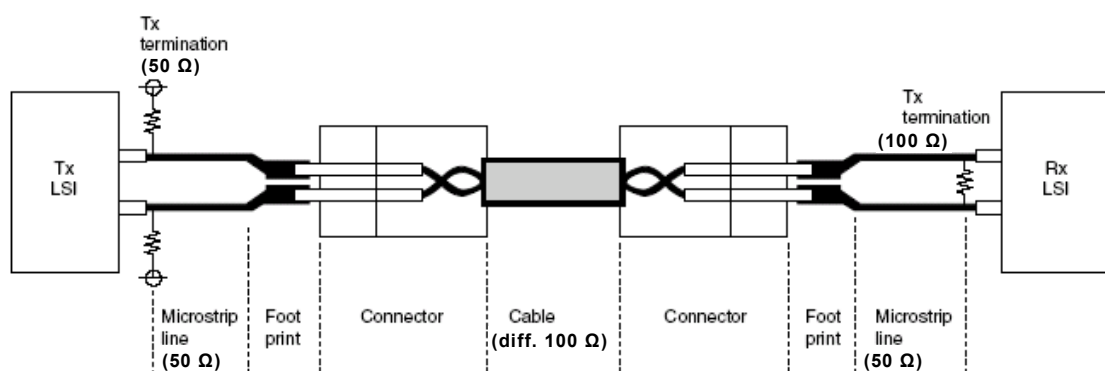
9 Transmission system and transmission line of electrical characteristics

The transmission systems (see Figure 9) are required to meet the specifications below.

- The differential impedance shall meet the specification stated in Figure 10. A transmission line has a small and gradual attenuation.
- A transmission line loss on a cable shall be less than –15 dB at 1 GHz in accordance with $\sqrt{f}$ attenuation. (See Figure 11).

The differential signal cable skew time shall be:

- less than 30 % of one bit time (SFTCLK > 33 MHz);
- less than 24 % of one bit time (SFTCLK ≤ 33 MHz).



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Figure 9 – Transmission system

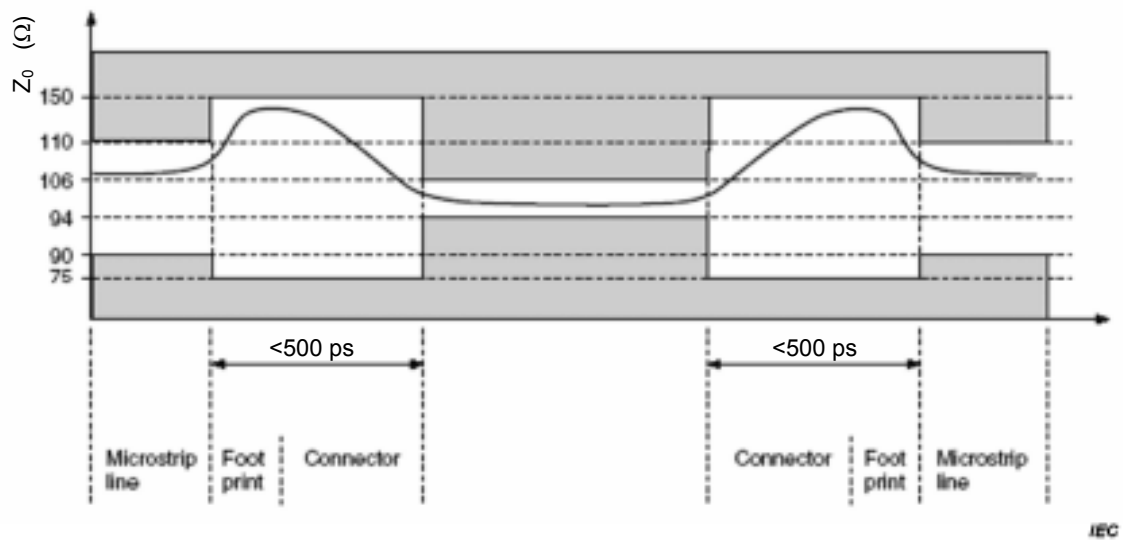


Figure 10 – Transmission line tolerance impedance

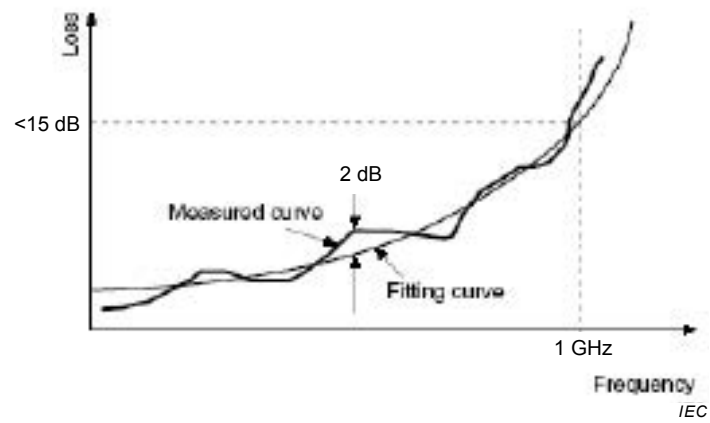


Figure 11 – Transmission loss

Annex A (informative)

Multiple link application

A.1 Single link application example

A.1.1 Block diagram for single link transmission

A block diagram of a differential single link is shown in Figure A.1.

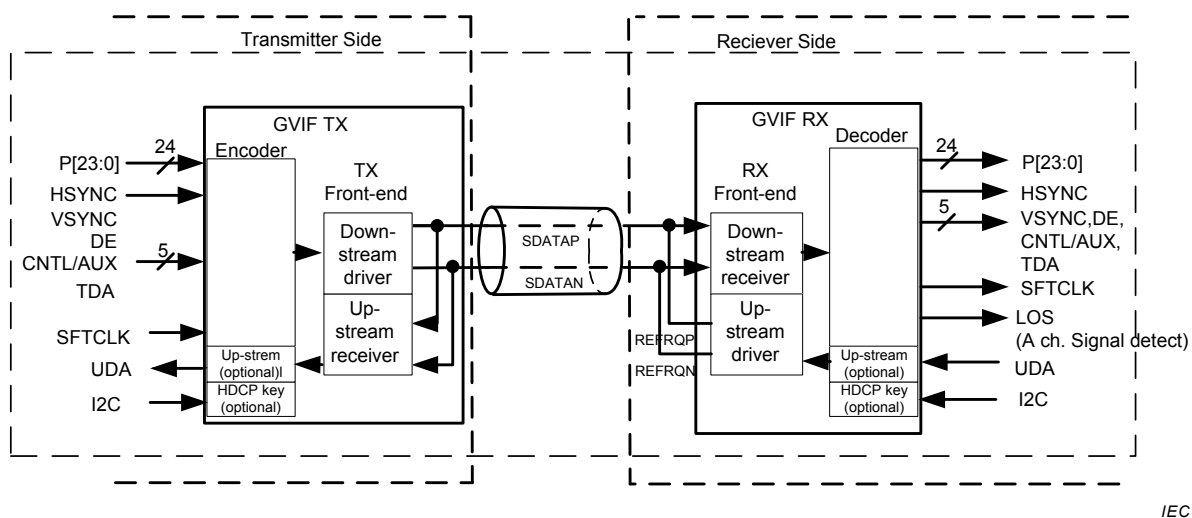


Figure A.1 – Differential single link block diagram

The down-stream transmission signal are the encoded serial P[23:0], HSYNC, VSYNC, DE, CNTL/AUX, SDA and TDA 30 bit data by the GVIF TX encoder. These data are in all of the SFTCLK domains, and they can be re-generated by the GVIF RX decoder. The encoder and decoder are performed in accordance with Clause 7.

The LOS signal output on the receiver side is asserted when GVIF RX receives no differential signal input or the clock and data recovery circuit in GVIF RX does not generate recovered SFTCLK (loss of lock).

There are optional functions for down-stream and up-stream user defined signal transmissions with terminal name CNTL/AUX and TDA for down-stream and UDA for up-stream as shown in Figure A.1. AUX can also optionally use an audio enable signal.

Additionally, HDCP (high-bandwidth digital contents protection) is also defined as an optional function. The I2C inputs for both GVIF TX and GVIF RX are input terminals to control the HDCP authentication function. The SDA (optional) is generated by an I2C signal order, i.e. a signal to exchange a HDCP key between GVIF TX and GVIF RX.

A.1.2 Data mapping of single link transmission

A data mapping array should be assigned pixel data of a LCD module as shown in Figure A.2.

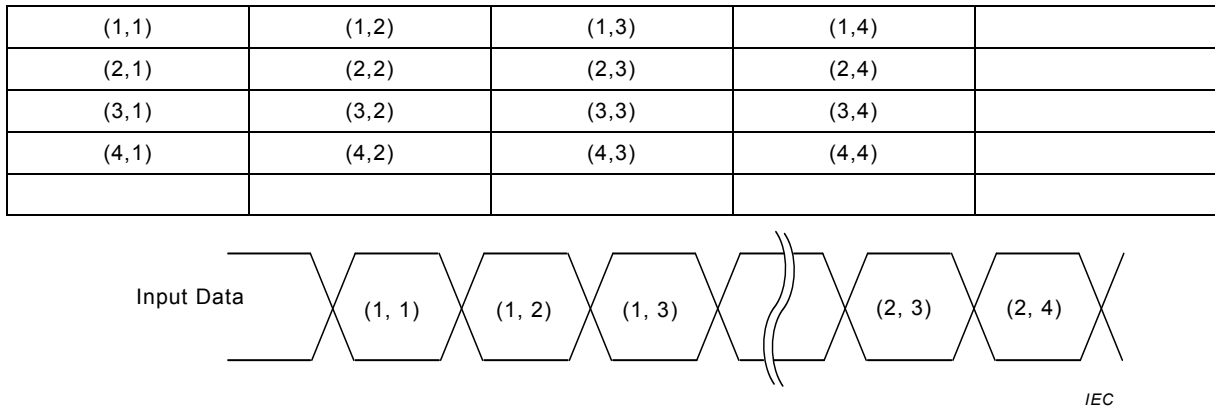


Figure A.2 – Pixel configuration

A.2 Multiple link application example

A.2.1 Block diagram for 2-pair parallel transmission

A block diagram of a multiple link system configuration is shown in Figure A.3. Each channel is called A-ch and B-ch.

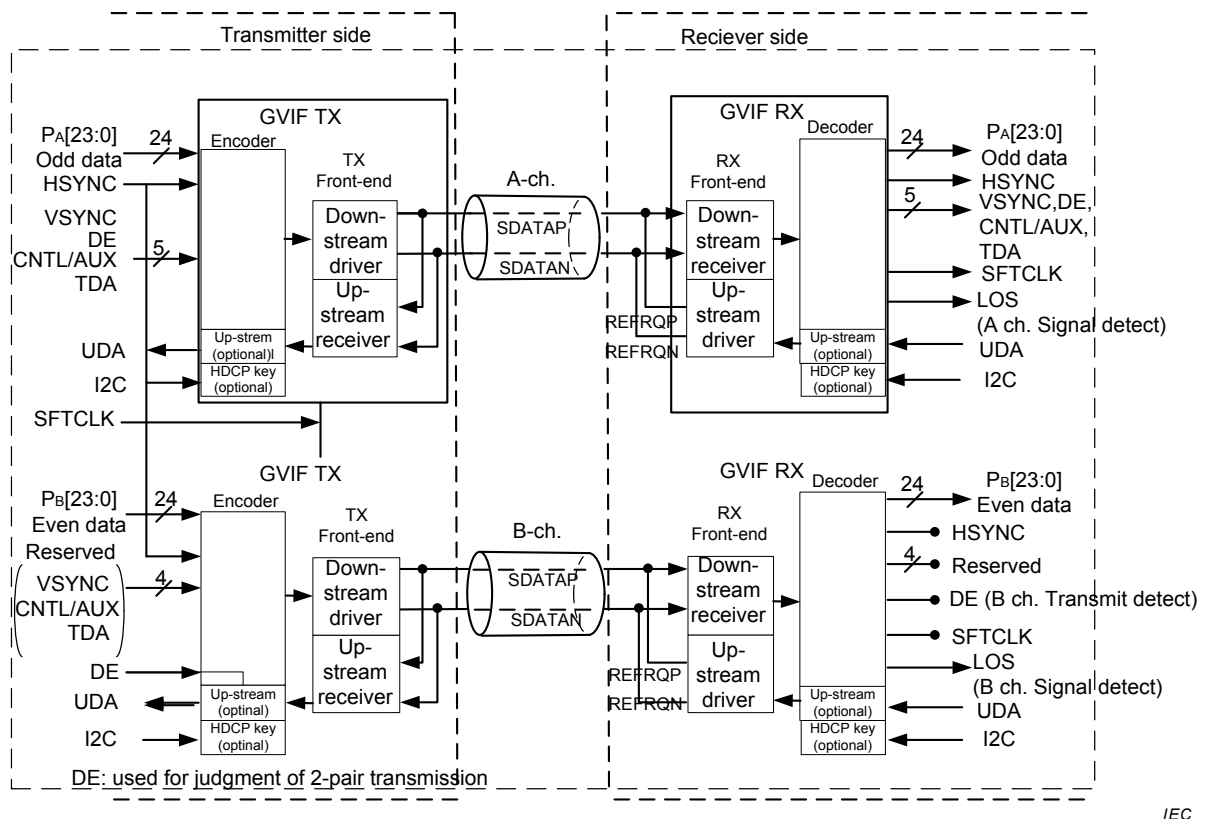


Figure A.3 – Multiple link application block diagram

In order to switch between a 1-pair transmission system and 2-pair transmission system, a detection circuit is necessary on the transmitter side. In case of a 2-pair transmission system,

the DE equivalent signal is input to the “DE” pin of B-ch, and in case of a 1-pair transmission system, the fixed signal is input to the “DE” pin of A-ch.

A.2.2 Data mapping of 2-pair transmission

Odd data should be assigned to A-ch, even data should be assigned to B-ch.

A data mapping array for a LCD module is shown in Figure A.4.

(1,1)	(1,2)	(1,3)	(1,4)	
(2,1)	(2,2)	(2,3)	(2,4)	
(3,1)	(3,2)	(3,3)	(3,4)	
(4,1)	(4,2)	(4,3)	(4,4)	
A-ch. data	B-ch. data	A-ch. data	B-ch. data	

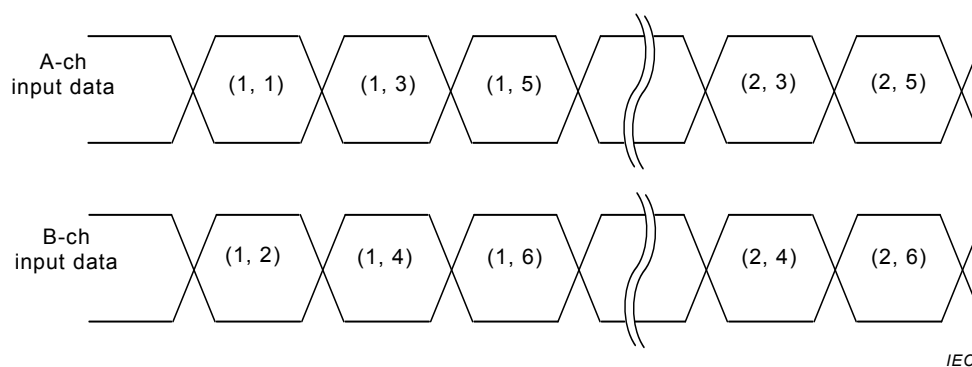


Figure A.4 – Pixel configuration when using 2-pairs

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